LMU · CHAIR OF AI & MACHINE LEARNING

Practical Course on shapiq

Project Topics — Summer Semester 2026

Eight projects. Pull requests + demos.
Real contributions to an actively-developed open-source library.

Max Muschalik · 23 April 2026 · Akademiestr. 7, R 103/105

01

PR + DEMO

Text & LLM Explanations

Explain what the model read.

Bring first-class text support to shapiq. Explain encoder classifiers and causal LLMs token-by-token, with honest benchmarks against SHAP, captum, and Inseq.

Core PR

A TextImputer (or shapiq.text package) — tokens / words / sentences × mask / pad / attention-mask × logit / next-token likelihood.

Effi ciency

Batched coalition eval, KV-cache for generation, PyTorch + JAX. Non-negotiable for LLM-scale models.

Demo

DistilBERT + a 1–8B causal LLM + seq2seq. Higher-order interactions (negation, entities, prompt injection).

02

PR(S)

Modern Explainer & Imputer Architecture

Clean foundations, measurable speedups.

Replace the ~150-line hand-rolled model-type match chain with lazy single-dispatch (flextype). Then add a batched-prediction protocol that makes neural backends actually fast.

Dispatch

Kill get_predict_function_and_model_type. Each backend registers its own handlers — torch/xgboost/lightgbm stay lazy imports.

Batched path

A predict_batched protocol on top of predict. torch DataLoader with pin_memory, device-aware, inference_mode.

Flax backend

Second-backend proof: add JAX/Flax in <50 lines. If it spills out of its module, the refactor isn't done.

DISPATCH REFACTOR

BEFORE

```
if model_class in sklearn...:
elif model_class in xgboost...:
elif model_class in lightgbm...:
elif ...   # ~150 lines
```

AFTER · flextype registry

```
@predict.register("torch.nn.Module")
@predict.register("flax.linen.Module")
@predict.register("xgboost.Booster")
```

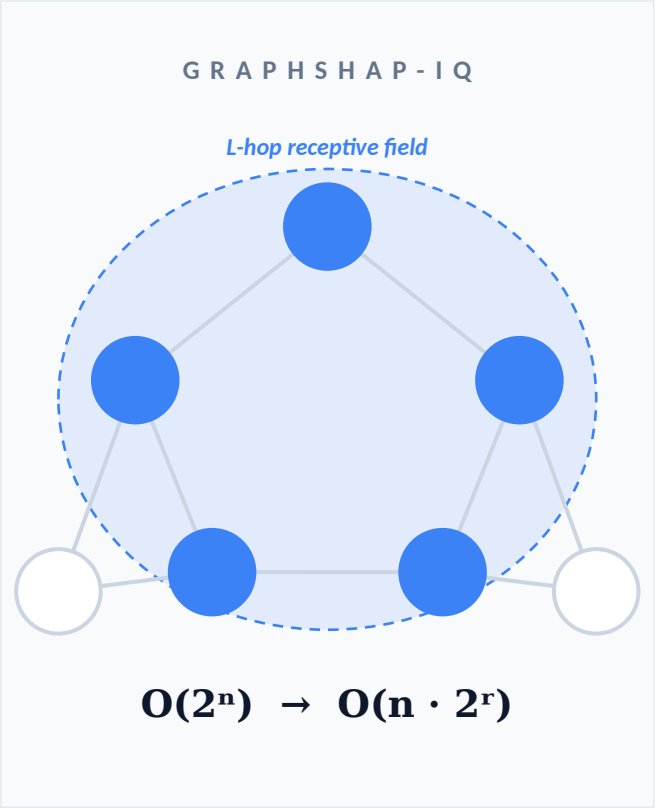
03

PR + DEMO

GraphSHAP-IQ — Shapley Interactions for GNNs

Exact, any-order, on graphs.

Exact Shapley interactions for Graph Neural Networks (ICLR '25). Exploit L-hop receptive fields to cut $O(2^n)$ to $O(n \cdot 2^r)$ — the graph analogue of TreeSHAP-IQ.



GraphGame

Subclass Game. Coalitions = node subsets, value = GNN prediction with absent nodes zeroed. Validate with ExactComputer.

Algorithm

Receptive-field BFS → Möbius transform → MoebiusConverter. Support GCN, GIN, GAT + max_interaction_size with efficiency redistribution.

Demo

Reproduce paper (MUTAG / AIDS), then apply to a new dataset. Compare GCN vs GIN vs GAT explanations.

04

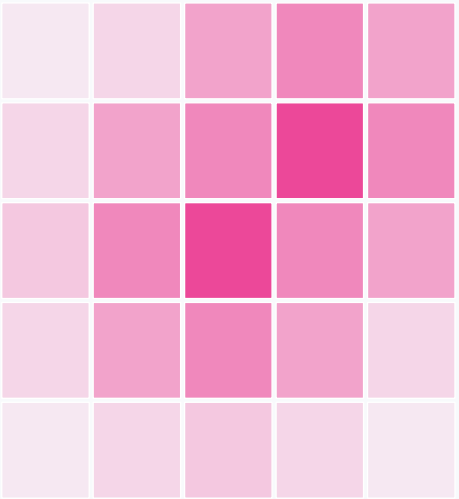
PR + DEMO

Image Explanations with shapiq

See what the model sees.

Right now shapiq has no real image support. Build a proper image imputer: pluggable players (patches / SLIC / custom masks) × masking strategies (mean / blur / attention / beyond).

IMAGE SHAPIQ



patches · superpixels · attention tokens

Imputer

Composable ImageImputer or new shapiq.image subpackage. PyTorch + JAX. Auto-defaults per model type (ViT → patches + attention; CNN → SLIC + mean).

Effi ciency

Batched coalitions, GPU-resident masking, configurable batch_size. Benchmark vs existing hard-coded games.

Demo

ViT-B/16, ResNet-50, one more. Reproduce FixLIP. Show pairwise interactions revealing context effects.

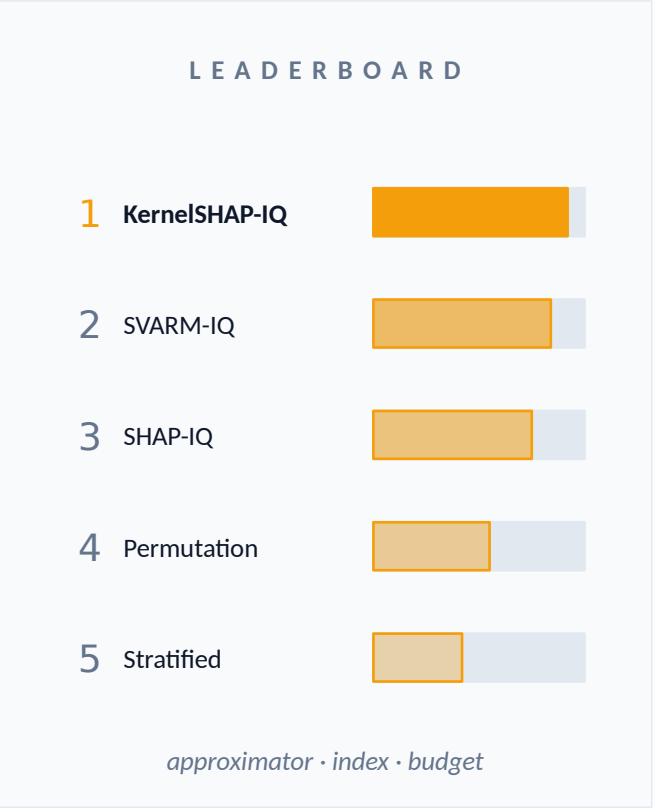
05

DEMO

Living Benchmark Leaderboard

One dashboard, every method.

Answer the question a shapiq user can't currently answer: "Which approximator should I use for my game, my budget, my index?" A polished Gradio app with rigorous, well-seeded benchmarks.



Runner

Build from scratch. GT from ExactComputer (n ≤ 14) or TreeSHAP-IQ/InterventionalTreeExplainer (any n) for tree games. Seed-robust, parallel.

Metrics

Think beyond raw MSE: normalized error, Spearman/Kendall, Precision@k. Let users switch metrics in the app.

App

Polished Gradio leaderboard: filter by game / index / budget. TabPFN HF Space is the UX north star.

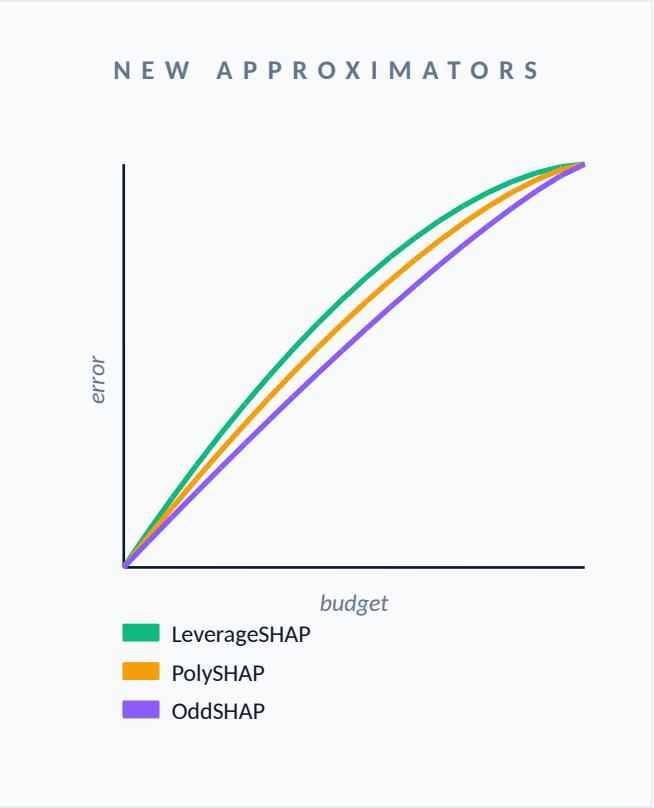
06

PR(S)

New Shapley Value Approximators

Push the approximator frontier.

Implement three recent approximators not yet in shapiq: LeverageSHAP, PolySHAP, OddSHAP. Validate against ExactComputer, reproduce paper experiments, benchmark against the existing line-up.



LeverageSHAP

Leverage-score sampling for KernelSHAP with near-optimal finite-sample error bounds (Musco & Witter, 2024).

PolySHAP

Polynomial-regression estimator for Shapley values.

OddSHAP

Variance reduction via odd-subset symmetry.

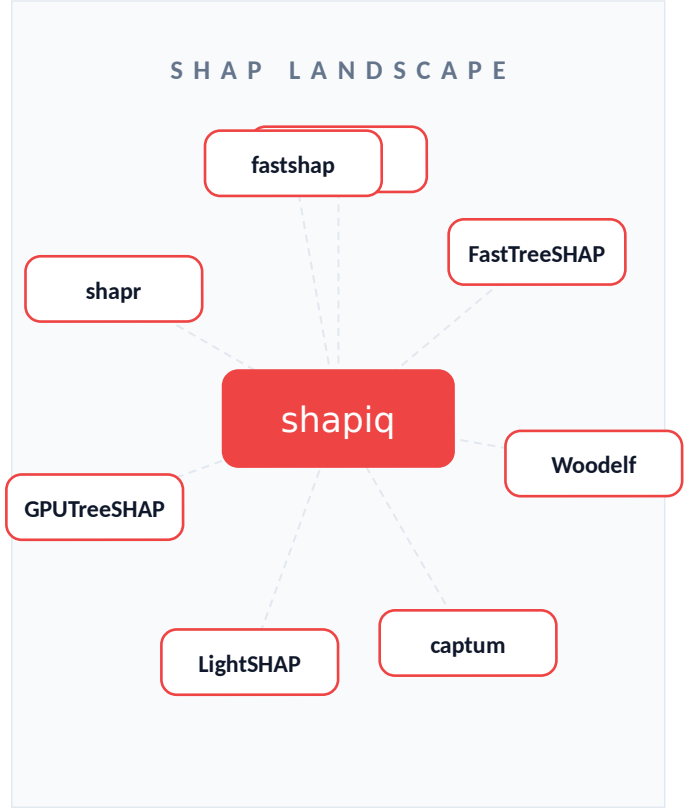
07

DEMO + PR(S)

The SHAP Landscape — shapiq vs the World

Know the field cold.

A thorough survey and head-to-head benchmark: shap, shapr, FastTreeSHAP, Woodelf, LightSHAP, captum, fastshap, GPUTreeSHAP, and more. Where does shapiq win? Where does it lose? Why?



Survey

Map the ecosystem. API, methods, model support, maintenance. Build the structured comparison table nobody has.

Benchmarks

TreeSHAP / KernelSHAP / PermutationSHAP head-to-heads on runtime, memory, and agreement. Where do 'same method' implementations differ?

Synthesis

Actionable recommendations — performance gaps, missing features, strategic positioning. Ship fixes as targeted PRs.

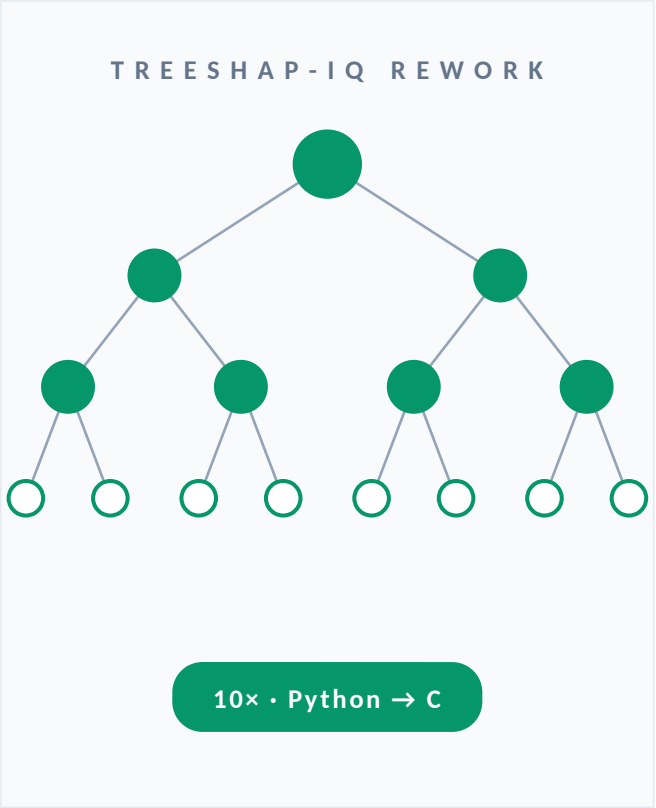
08

PR(S)

TreeSHAP-IQ Rework — C Extension & BII

Faster, more indices, fewer bugs.

TreeSHAP-IQ is 770 lines of Python polynomial arithmetic with known dead code and suspected bugs. Clean it up, fix it, extend it to BII/BV, and port the hot path to C.



Cleanup

Remove the unreachable STII/FSII branches. Audit and validate the existing SV/SII/k-SII paths against ExactComputer — don't trust the current code.

BII & BV

Route BII/BV through the CII path (the weights are already half there). Validate at multiple interaction orders against ExactComputer and TreeSHAPIQXAI.

C extension

Port the algorithm to C/C++ using interventional/cext/ as a reference. Aim 10×+ speedup, identical results to the Python fallback.

NEXT UP

Form your group. Pick your project.

Today — 10:30	**Group forming**	Right here in this room.
Tomorrow — noon	**Project pick + group name**	Mail to maximilian.muschalik@ifi.lmu.de.
Tomorrow — EoD	**Groups + topics announced**	Check Moodle.
29 April	**First group meetings**	Bring slides. 45 min each.
10 July	**Final submission**	PRs merged. Demos polished.

Questions? Discussion forum · Kostas (env/setup) · Max (everything else)